Theory of Logic: Proposition, Truth Tables, Tautology, Satisfiability, Contradiction, Algebra of proposition, Theory of Inference,

Predicate Logic: First order predicate, well formed formula of predicate, quantifiers, Inference theory of predicate logic.

Theory of Logic

Theory of logic, also known as Mathematical Logic, is the study of logical statements, arguments and inference.

Branches of Logic:

1. Propositional Logic: Deals with statements that can be true or false.
2. Predicate Logic: Extends propositional logic to include predicates and quantifiers.
3. Modal Logic: Concerned with possibility, necessity and obligation.
4. Description Logic: Combines logic and ontology for knowledge representation.
5. Temporal Logic: Studies time and temporal relationship.

Proposition: A proposition or statement is a declarative sentence that is either true or false.

Example: (1) "Three plus three equals six."
(2) "Three plus three equals seven."
Both are statements but first is true and second is false.
(3) "$x+y>1$" is not a statement because for some values of x and y it is true and for others it is false.

The truth or falsity of a statement is called its **Truth value.**

Since only two possible truth values, therefore this logic is also called two-valued logic.

More Examples

	Truth Value	Is Proposition?
1. The sun raises in the east.	TRUE	YES
2. $2+4=6$	TRUE	YES
3. The sun rises in the west.	False	YES
4. Do you speak Hindi?	Not a declarative	NO
5. $4-x=8$	Declarative but may be both TRUE or FALSE	NO
6. Close the door.	Command	NO
7. What a hot day!	Exclamation	NO
8. We shall have milk for dinner.	Either TRUE or FALSE, but not both	YES

Propositional variables : $p, q, r, s, \ldots\ldots$
Propositional constants : T, F

Compound Proposition

A proposition consisting of only a single propositional variable or a single propositional constant is called an atomic (primary, primitive) proposition or simply proposition. that is they can not be further subdivided.

A proposition obtained from the combinations of two or more propositions by means of logical operators or connectives of two or more propositions or by negating a single proposition is referred to molecular or composite or compound proposition.

Connectives

(1) Negation: If p is any proposition, the negation of p is denoted by $\sim p$ or $\neg p$. $\sim p$ read as not p.
when p is true, $\sim p$ is false.
p is false, $\sim p$ is true.

Let p: Paris is in France.
Then the negation of p is the statement,
$\sim p$: Paris is not in France.

Important

The following propositions all have the same meaning:

p: All people are intelligent.
q: Every person is intelligent.
r: Each person is intelligent.
s: Any person is intelligent.

Find negation of following:

(1) p: All students are intelligent.

$\sim p$: Some students are not intelligent

OR

$\sim p$: There exist a student who is not intelligent.

OR

$\sim p$: At least one student is not intelligent.

(2) q: No student is intelligent

$\sim q$: Some students is intelligent.

Imp Note: "No student is intelligent" is not negation of p.
"All students are intelligent" is not negation of q.

(2) Conjunction: $p \wedge q$

Example: p: Ram is healthy.
q: He has blue eyes.

$p \wedge q$: Ram is healthy and he has blue eyes.

(3) Disjunction: $p \vee q$

Example: p: It is cold. q: It is raining.

$p \vee q$: It is cold or raining.

(4) Conditional: $p \rightarrow q$

(5) Biconditional: $p \leftrightarrow q$

Symbol Used	Connective Word	Nature	Symbolic Form	Negation
$\sim, \neg$	not	Negation	$\sim p$	$\sim(\sim p) \equiv p$
$\wedge$	and	Conjunction	$p \wedge q$	$\sim(p \wedge q) \equiv (\sim p) \vee (\sim q)$
$\vee$	or	Disjunction	$p \vee q$	$\sim(p \vee q) \equiv (\sim p) \wedge (\sim q)$
$\Rightarrow, \rightarrow$	if .... then	Implication or conditional	$p \rightarrow q$ $\equiv (\sim p) \vee q$	$\sim(p \rightarrow q) \equiv \sim(\sim p \vee q)$ $\equiv p \wedge (\sim q)$
$\Leftrightarrow, \leftrightarrow$	if and only if	Equivalence or Biconditional	$p \Leftrightarrow q$ $\equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$ $\equiv (\sim p \vee q) \wedge (\sim q \vee p)$	$\sim(p \Leftrightarrow q)$ $\equiv (p \wedge (\sim q)) \vee (q \wedge (\sim p))$

Truth Table: A truth tables shows truth values of a compound proposition for all possible combinations of truth values of atomic propositions in compound proposition.

p	$\sim p$
F	T
T	F

p	q	$p \vee q$
F	F	F
F	T	T
T	F	T
T	T	T

p	q	$p \wedge q$
F	F	F
F	T	F
T	F	F
T	T	T

p	q	$p \rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

p	q	$p \leftrightarrow q$
F	F	T
F	T	F
T	F	F
T	T	T

Law of Duality: Two statements p and q are said to be dual of each other if either one can be obtained from the other by replacing $\wedge$ by $\vee$, $\vee$ by $\wedge$ and F by T, T by F.

Ex→ Dual of $(p \vee q) \wedge r$ is $(p \wedge q) \vee r$

Dual of $(p \wedge q) \vee T$ is $(p \vee q) \wedge F$

Principle of Duality: If any two statements are equal then their dual are also equal.

Q: Construct a truth table for each compound statement.

(i) $p \wedge (\sim q \vee q)$ (ii) $\sim(p \vee q) \vee (\sim p \wedge \sim q)$

Ans: (i)

p	q	$\sim q$	$\sim q \vee q$	$p \wedge (\sim q \vee q)$
F	F	T	T	F
F	T	F	T	F
T	F	T	T	T
T	T	F	T	T

(ii) Do yourself.

Tautology, Contradiction, Satisfiability (Contingency)

- A compound proposition is called a Tautology if it's truth value is always TRUE for all possible combination of truth values of its variables.
- A compound proposition is called a contradiction, if it's truth value is always FALSE.
- A proposition that is neither a tautology nor a contradiction is called a contingency.

Example: (i) $p \vee \sim p$ (ii) $\sim(p \wedge q) \vee q$ (iii) $p \rightarrow (p \vee q)$ are tautology.

(iii)

p	q	$p \vee q$	$p \rightarrow (p \vee q)$
F	F	F	T
F	T	T	T
T	F	T	T
T	T	T	T

$\because p \rightarrow (p \vee q)$ is always TRUE

$\therefore p \rightarrow (p \vee q)$ is a Tautology

Example: $p \wedge (q \wedge \sim p)$ is a contradiction.

Example: $p \vee q$, $p \wedge q$, $p \rightarrow q$, $p \leftrightarrow q$ are contingency.

Logical Equivalence

If two propositions $P(p,q,\cdots)$ and $Q(p,q,\cdots)$ where $p,q,\cdots$ are propositional variables have the same truth values in every possible case, the propositions are called logically equivalent or simply equivalent, and denoted by

$$P(p,q,\cdots) \equiv Q(p,q,\cdots)$$

How to test?
Construct truth Table for P and Q, if in each row truth values for P and Q are same, then $P \equiv Q$.

Example: $p \to q \equiv \sim p \vee q$

p	q	$\sim p$	$p \to q$	$\sim p \vee q$
F	F	T	T	T
F	T	T	T	T
T	F	F	F	F
T	T	F	T	T

Identical

$\therefore \quad p \to q \equiv \sim p \vee q$

Converse, Inverse, Contrapositive:

p	q	Conditional $p \to q$	Converse $q \to p$	Inverse $\sim p \to \sim q$	contrapositive $\sim q \to \sim p$
F	F	T	T	T	T
F	T	T	F	F	T
T	F	F	T	T	F
T	T	T	T	T	T

Note: Converse: $q \to p$ Inverse: $\sim p \to \sim q$ Contrapositive: $\sim q \to \sim p$ are of given $p \to q$

Example: p: It rains, q: The crops will grows
$p \to q$: If it rains then the crops will grow.

Converse of $p \to q$ is $q \to p$: If the crops grows, then there has been rain.
Inverse of $p \to q$ is $\sim p \to \sim q$: If it does not rain then the crops will not grow.
Contrapositive of $p \to q$ is $\sim q \to \sim p$: If the crops do not grow there has been no rain.

Algebra of Proposition:

1. Idempotent laws
   a. $p \vee p \equiv p$ b. $p \wedge p \equiv p$
2. Associative laws
   a. $(p \vee q) \vee r \equiv p \vee (q \vee r)$ b. $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
3. Commutative laws
   a. $p \vee q \equiv q \vee p$ b. $p \wedge q \equiv q \wedge p$
4. Distributive laws
   a. $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ b. $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
5. Identity laws
   a. $p \vee F \equiv p$ b. $p \wedge T \equiv p$
   c. $p \vee T \equiv T$ d. $p \wedge F \equiv F$
6. Complement laws
   a. $p \vee \sim p \equiv T$ b. $p \wedge \sim p \equiv F$
   c. $\sim T \equiv F$ d. $\sim F \equiv T$
7. Involution law
   $\sim(\sim p) \equiv p$
8. Demorgan's laws
   a. $\sim(p \vee q) \equiv \sim p \wedge \sim q$ b. $\sim(p \wedge q) \equiv \sim p \vee \sim q$
9. Absorption laws
   a. $p \vee (p \wedge q) \equiv p$ b. $p \wedge (p \vee q) \equiv p$

a. Proof: $p \vee (p \wedge q) \equiv (p \wedge T) \vee (p \wedge q)$
$\equiv p \wedge (T \vee q)$ [reverse of distribution laws]
$\equiv p \wedge T$ Identity laws
$\equiv p$ Identity laws

Disjunctive Normal Form (DNF)

Elementary products: $p \wedge \sim q$, $\sim p \wedge \sim q$, $\sim p \wedge q$, $p \wedge q$, $p \wedge \sim p$, $p \wedge p$, etc.

Elementary Sums: $p \vee q$, $p \vee \sim q$, $\sim p \vee \sim q$ etc.

A logical expression is said to be in DNF, if it is the sum of elementary products.

Example: $p \vee (q \wedge r)$ and $p \vee (\sim q \wedge r)$

$\because$ $p \vee (q \wedge r) \equiv (p \wedge p) \vee (q \wedge r)$

$p \vee (\sim q \wedge r) \equiv (p \wedge p) \vee (\sim q \wedge r)$

Procedure to obtain a DNF:

1. Remove all $\rightarrow$ and $\leftrightarrow$ by an equivalent containing $\wedge, \vee, \sim$ only
2. Eliminate $\sim$ before sums and products
3. Apply distributive law

Example

(a) $p \wedge (p \rightarrow q) \equiv p \wedge (\sim p \vee q)$

$\equiv (p \wedge \sim p) \vee (p \wedge q)$

Note: DNF of given logical expression is not unique.

Conjunctive Normal Form (CNF)

A logical expression is said to be in CNF if it consists of a product of elementary sum.

Example: $(p \vee q) \wedge (p \vee \sim q)$, $p \wedge (q \vee r)$

Q. Obtain CNF of $p \wedge (p \rightarrow q)$.

Ans: $p \wedge (p \rightarrow q) \equiv p \wedge (\sim p \vee q)$

Principal DNF (PDNF)

It contains only minterms.

Minterms of two variables p and q are

$p \wedge q$, $p \wedge \sim q$, $\sim p \wedge q$, $\sim p \wedge \sim q$ $\boxed{2^n}$

PDNF form of a given formula can be defined as an equivalent formula consisting of disjunctives of minterms only. This is also called $\boxed{SOP}$

The process of obtaining pdnf:

(By Truth Table Method)

Example $p \Rightarrow q$

p	q	$p \Rightarrow q$	
F	F	T ✓	$(\sim p \wedge \sim q)$
F	T	T ✓	$(\sim p \wedge q)$
T	F	F	
T	T	T ✓	$(p \wedge q)$

$\therefore p \Rightarrow q \equiv (\sim p \wedge \sim q) \vee (\sim p \wedge q) \vee (p \wedge q)$

PCNF It contains only maxterms. $\boxed{POS}$

Maxterms of two variables p and q are

$p \vee q$, $p \vee \sim q$, $\sim p \vee q$, $\sim p \vee \sim q$

Example: obtain pcnf of $p \wedge q$.

p	q	$p \wedge q$	
F	F	F ✓	$(p \vee q)$
F	T	F ✓	$(p \vee \sim q)$
T	F	F ✓	$(\sim p \vee q)$
T	T	T	

$\therefore p \wedge q \equiv (p \vee q) \wedge (p \vee \sim q) \wedge (\sim p \vee q)$

Well formed formula (wff)

A statement formula is a string consisting of variables, parentheses and connectives symbol. A statement formula is called well formed formula. "A wff is a syntactically correct expression constructed from atomic proposition, connectives & parentheses"

(i) A statement variable p standing alone is a well formed formula.

(ii) If p is a well formed formula, then $(\sim p)$ is a well formed formula.

(iii) If p and q are well formed formula then $(p \wedge q)$, $(p \vee q)$, $(p \Rightarrow q)$ and $(p \Leftrightarrow q)$ are wffs.

(iv) A string of symbols is wff if and only if it is obtained by infinitely many applications of the rule (i), (ii) and (iii).

Valid Argument

An argument $p_1, p_2, \ldots, p_n \vdash q$ is called valid if q is true whenever all its premises $p_1, p_2, \ldots, p_n$ are true.

An argument is called valid if the premises implies the conclusion.

Thus the argument $p_1, p_2, \ldots, p_n \vdash q$ is said to be valid if and only if the statement

$$(p_1 \wedge p_2 \wedge \ldots \wedge p_n) \rightarrow q$$ is a tautology.

An argument which is not valid is said to be a fallacy or an invalid Argument.

$$\left.\begin{array}{l} p_1 \\ p_2 \\ \vdots \\ p_n \end{array}\right\} \text{premises}$$

$$\underline{\quad q \quad} \} \text{ conclusion}$$

Q. Show that the following argument is valid:

$$\begin{array}{l} p \vee q \\ \sim p \\ \hline \therefore\ q \end{array}$$

Ans:

p	q	$\sim p$	$p \vee q$	$(p \vee q) \wedge (\sim p)$	$[(p \vee q) \wedge (\sim p) \rightarrow q]$
F	F	T	F	F	T
F	T	T	T	T	T
T	F	F	T	F	T
T	T	F	T	F	T

$\because$ $[(p \vee q) \wedge (\sim p) \rightarrow q]$ is a tautology

$\therefore$ Argument is valid.

Alternate method

$p \rightarrow q$ } Premises
$q \rightarrow r$ }

$p \rightarrow r$ } conclusion

premises: $p \rightarrow q$, $q \rightarrow r$; conclusion: $p \rightarrow r$

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$
F	F	F	T	T	T
F	F	T	T	T	T
F	T	F	T	F	T
F	T	T	T	T	T
T	F	F	F	T	F
T	F	T	F	T	T
T	T	F	T	F	F
T	T	T	T	T	T

Step (i) Identify critical rows (all premises are true).
(ii) If conclusion is also true in all critical rows, then argument is valid.

$\because$ Rows 1, 2, 5 & 8 are critical rows and conclusions also TRUE. Hence argument is valid.

Rules of Inference

Rule of Inference	Tautological form	Name
1. $\dfrac{p}{\therefore p \vee q}$	$p \to (p \vee q)$	Addition
2. $\dfrac{p \wedge q}{\therefore p}$	$(p \wedge q) \to p$	Simplification
	$((p) \wedge (q)) \to (p \wedge q)$	Conjunction
3. $\dfrac{\begin{matrix}p\\q\end{matrix}}{\therefore p \wedge q}$	$[(p \to q) \wedge p] \to q$	Modus Ponens (Rule of detachment) Fundamental Rule 1
4. $\dfrac{\begin{matrix}p \to q\\p\end{matrix}}{\therefore q}$		
5. $\dfrac{\begin{matrix}p \to q\\\sim q\end{matrix}}{\therefore \sim p}$	$[(p \to q) \wedge (\sim q)] \to (\sim p)$	Modus tollens
6. $\dfrac{\begin{matrix}p \to q\\q \to r\end{matrix}}{\therefore p \to r}$	$[(p \to q) \wedge (q \to r)] \to (p \to r)$	Hypothetical Syllogism Fundamental Rule 2
7. $\dfrac{\begin{matrix}p \vee q\\\sim p\end{matrix}}{\therefore q}$	$[(p \vee q) \wedge (\sim p)] \to q$	Disjunction Syllogism
8. $\dfrac{\begin{matrix}(p \to q) \wedge (r \to s)\\p \wedge r\end{matrix}}{\therefore q \vee s}$	$[[(p \to q) \wedge (r \to s)] \wedge (p \wedge r)] \to (q \vee s)$	Constructive dilemma
9. $\dfrac{\begin{matrix}(p \to q) \wedge (r \to s)\\\sim q \vee \sim s\end{matrix}}{\therefore \sim p \vee \sim r}$	$[(p \to q) \wedge (r \to s) \wedge (\sim q \vee \sim s)] \to (\sim p \vee \sim r)$	Destructive dilemma

Q. Using rule of inference, determine whether the following inference patterns are valid or not.

$\sim t \rightarrow \sim r$
$\sim s$
$t \rightarrow w$
$r \vee s$

$\therefore w$

Solution:

1.	$r \vee s$	given
2.	$s \vee r$	commutative law
3.	$\sim s$	given
4.	r	Disjunction syllogism using 2 & 3
5.	$\sim t \rightarrow \sim r$	given
6.	$\sim t$	Modus tollen using 5 & 4
7.	$t \rightarrow w$	given
8.	w	Modus ponens using 7 & 6

Hence the given Inference pattern is valid.

Q. Show that t is valid conclusion from the premises $p \rightarrow q$, $q \rightarrow r$, $r \rightarrow s$, $\sim s$ and $p \vee t$.

Solution: Given

$p \rightarrow q$
$q \rightarrow r$
$r \rightarrow s$
$\sim s$
$p \vee t$

$\therefore t$

premises

} Conclusion

1.	$p \rightarrow q$	given
2.	$q \rightarrow r$	given
3.	$p \rightarrow r$	Hypothetical syllogism using 1 & 2
4.	$r \rightarrow s$	given
5.	$p \rightarrow s$	Hypothetical syllogism using 3 & 4
6.	$\sim s$	given
7.	$\sim p$	Modus tollens using 5 & 6
8.	$p \vee t$	given
9.	t	Disjunctive syllogism using 7 & 8

Thus we can conclude that t is valid conclusion from given premises.

Predicate Logic (calculus)

Predicate calculus is a generalization of propositional calculus. It contains all the components of propositional calculus, including propositional variables and constants.

Predicate calculus is important for several reasons, this has application in expert system, in database and also basis for the Prolog language.

A part of a proposition describing the properties of an object or relation among objects is called a predicate. For example:

Ram is a good boy.
Shym is a good boy.

The two propositions can be replaced by a single proposition "x is a good boy".

Both Ram and Shyam has the same property of being good boy.

here, phrase or predicate "is a good boy"

A Predicate is symbolized by capital letters and the names of objects or individuals in general by small letters.

The sentence "x is a good boy" is symbolised as P(x), where x is a predicate variable and P is predicate.

When concrete values are substituted in place of x (predicate variable), a statement results.

P(x) is also called a propositional function.

Each choice of x produces a proposition P(x) that is either TRUE or False.

Universal and Existential quantifier

The phrase "for all" denoted by $\forall$ is called the universal quantifier.

Example:
"All human beings are mortal"

Let $P(x)$: x is mortal
Then the above sentence can be written as
$\forall x\ P(x)$

for all x
for every x
for each x
} are same.

$\forall x\ P(x)$ is called universal statement.

$\forall x\ P(x)$ is true if $P(x)$ is true for every x.
$\forall x\ P(x)$ is false, if and only if $P(x)$ is false for at least one x.

The Phrase "there exists" denoted by $\exists$ is called the existential quantifier.

Example:
"there exists x such that $x^2=5$"
can be written as $\exists x\ P(x)$
where $P(x)$: $x^2=5$.

There exist an x
There is an x
For some x
There is at least one x
} are same

$\exists x\ P(x)$ is True if $P(x)$ is true for at least one x.
$\exists x\ P(x)$ is false if $P(x)$ is false for every x.

First order Predicate logic (FOPL)

FOPL extends propositional logic by introducing predicates, variables and quantifiers.

Rules of Formation:

1. Atomic formulas (e.g. $P(x)$) are FOPL formulas.
2. If $P(x)$ is a FOPL, then $\sim P(x)$ is also a FOPL formula.
3. If $P(x)$ and $Q(x)$ are FOPL formulas then

$$\left.\begin{array}{l} P(x) \wedge Q(x) \\ P(x) \vee Q(x) \\ P(x) \rightarrow Q(x) \\ \forall x\, P(x) \\ \exists x\, P(x) \end{array}\right\}$$ are also FOPL formulas.

Well formed formula of Predicate

A well formed formula of predicate is a syntactically correct expression constructed from predicate, variables, constants, connectives, quantifiers and parentheses.

Examples of WFFs:

$\forall x\, P(x)$

$\exists x\, (P(x) \wedge Q(x))$

$(\forall x)(\forall y)(R(x,y) \rightarrow R(y,x))$

$(\forall x)(P(x) \rightarrow Q(x))$

Examples of Non-WFFs:

$P(x$ missing right parenthesis.

$\forall x\, (P(x)$

$\exists x, y\, (P(x,y))$

After the correction above will be

$P(x)$

$(\forall x)\, P(x)$ or $\forall x\, P(x)$

$\exists x \forall y\, (P(x,y))$ or $\exists x \exists y\, P(x,y)$

Translating Sentences into Logical Expressions

1. Consider, Every person is precious.

Let $M(x)$: x is person.
$P(x)$: x is precious.

NOTE: for EVERY, use $\forall$ ($\rightarrow$)

$\therefore (\forall x)(M(x) \rightarrow P(x))$

2. Some student of this college passed GATE examination.

$C(x)$: x is student of this college.
$G(x)$: x passed GATE examination.

Note: for, some, use $\exists$ ($\wedge$)

$\therefore (\exists x)(C(x) \wedge G(x))$

Example: Let $K(x)$: x is man, $L(x)$: x is mortal
$M(x)$: x is an integer
$N(x)$: x is either positive or negative.

(a) All men are mortal. $(\forall x)(K(x) \rightarrow L(x))$
(b) Any integer is either positive or negative. $(\forall x)(M(x) \rightarrow N(x))$

Example: Let $K(x)$: x is student, $M(x)$: x is clever.
$N(x)$: x is successful.

(a) There exist a student. $(\exists x)(K(x))$
(b) Some students are clever. $(\exists x)(K(x) \wedge M(x))$
(c) Some student are not successful. $(\exists x)(K(x) \wedge \sim N(x))$

Negation of Quantified statements

Let $P(x)$: x is the student of the class has taken a course in discrete mathematics

"All students in the class have taken a course in discrete mathematics." $\forall x\, P(x)$

It's negation read as:

"There is at least one student in the class has not taken course in discrete mathematics." $\exists x \sim P(x)$

$\therefore \boxed{\sim \forall x\, P(x) \equiv \exists x \sim P(x)}$

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Similaly

$$\sim \exists x\, P(x) \equiv \forall x \sim P(x)$$

$\exists x\, P(x)$ interprete there is at least one x for which $P(x)$ is TRUE

Negation of $\exists x\, P(x)$ { Now; when $\exists x P(x)$ will be false?
When there is no x for which $P(x)$ is TRUE.

i.e. $(\forall x)(\sim P(x))$

$$\therefore \boxed{(\sim \exists x)(P(x)) \equiv (\forall x)(\sim P(x))}$$

Rules of Inference in Predicate logic

1. Universal Instantiation:

$$\frac{\forall x\, P(x)}{\therefore P(c)}$$ c is some element of the universe.

2. Existential instantiation:

$$\frac{\exists x\, P(x)}{\therefore P(c)}$$ c is some element for which $P(c)$ is TRUE.

3. Universal Generalization:

$$\frac{P(x)}{\therefore \forall x\, P(x)}$$

4. Existential Generalization:

$$\frac{P(c)}{\therefore \exists x\, P(x)}$$ c is some element of universe.

5. **Universal Modus Ponens**

$$\forall x\,(P(x) \rightarrow Q(x))$$
$$P(a)$$
$$\therefore Q(a)$$

6. **Universal Modus Tollens**

$$\forall x\,(P(x) \rightarrow Q(x))$$
$$\sim Q(a)$$
$$\therefore \sim P(a)$$

Demorgan's law for Quantified statement

(i) $\sim(P(x) \vee Q(x)) \equiv \sim P(x) \wedge \sim Q(x)$

(ii) $\sim(P(x) \wedge Q(x)) \equiv \sim P(x) \vee \sim Q(x)$

Use (i) & (ii) as given following in quantified statement:

$$\sim \forall x\,(P(x) \vee Q(x)) \equiv \exists x \sim(P(x) \vee Q(x)) \equiv \exists x(\sim P(x) \wedge \sim Q(x))$$

$$\sim \exists x\,(P(x) \vee Q(x)) \equiv \forall x \sim(P(x) \vee Q(x)) \equiv \forall x\,(\sim P(x) \wedge \sim Q(x))$$

etc.

Multiple Quantifiers: Let $P(x,y)$: x admires y. Then

$(\forall x)(\forall y)\,P(x,y)$: Every one admires everybody.
$(\forall x)(\exists y)\,P(x,y)$: Every one admires someone.
$(\exists x)(\forall y)\,P(x,y)$: Some one admires everyone.
$(\exists x)(\exists y)\,P(x,y)$: Some one admires somebody.
$(\forall y)(\forall x)\,P(x,y)$: Everybody is admired by everyone.
$(\forall y)(\exists x)\,P(x,y)$: Everybody is admired by someone.
$(\exists y)(\forall x)\,P(x,y)$: There is someone whom everyone admires.
$(\exists y)(\exists x)\,P(x,y)$: There is someone whom some one admires.

Important Questions

Q.(1). Test the validity of the following argument:
"If there was a ball game, then travelling was difficult.
If they arrived ontime, then travelling was not difficult.
They arrived on time. Therefore, there was no ball game."

Ans: Let
p: there was a ball game.
q: travelling was difficult.
r: they arrived on time

Symbolically, the argument is

$p \to q$
$r \to \sim q$
r
premises

$\therefore \sim p$ — conclusion

Proof:

1.	$r \to \sim q$	(given)
2.	r	given
3.	$\sim q$	from 1 & 2
4.	$p \to q$	given
5.	$\sim p$	from 3 & 4

$\therefore$ Argument is valid.

Q.(2). Show the validity of the following argument:
Hypotheses: "It is not sunny this afternoon, and it is colder than yesterday. We will go swimming only if it is sunny. If we do not go swimming, then we will take a canoe trip. If we take a canoe trip, then we will be home by sunset." Conclusion: "We will be home by sunset."

Let p: It is sunny this afternoon.
q: It is colder than yesterday.
r: We will go swimming.
s: We will take a canoe trip.
t: We will be home by sunset.

So, argument is

$\sim p \wedge q$
$r \to p$
$\sim r \to s$
$s \to t$
$\therefore t$

Proof:

1.	$\sim p \wedge q$	given
2.	$\sim p$	from 1
3.	$r \to p$	given
4.	$\sim r$	from 2 & 3
5.	$\sim r \to s$	given
6.	s	from 4 & 5
7.	$s \to t$	given
8.	t	from 6 & 7

$\therefore$ given argument is valid.